

Practical 4

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4.1.1 Pandas – Series Creation and Manipulation

Problem Statement:

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Code:

```
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
series = pd.Series(numbers)

# Grouping by even and odd numbers and calculating the mean
grouped = series.groupby(series % 2 == 0).mean()

# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
print("Mean of even and odd numbers:")
print(grouped)
```

Output:

Average time
0.053 s
53.17 ms

Maximum time
0.074 s
74.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 150 msDebug

Expected output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd ... 5.0

Even ... 6.0

dtype: float64

Actual output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd ... 5.0

Even ... 6.0

dtype: float64

Test case 274 msDebug

Expected output

4 4 4 6 6 2 2 2 10 12 14 16

Mean of even and odd numbers:

Even ... 6.833333

dtype: float64

Actual output

4 4 4 6 6 2 2 2 10 12 14 16

Mean of even and odd numbers:

Even ... 6.833333

dtype: float64

Test case 368 msDebug

Expected output

1 5 7 9 11 13 15 17 111 21 25

Mean of even and odd numbers:

Odd ... 21.363636

dtype: float64

Actual output

1 5 7 9 11 13 15 17 111 21 25

Mean of even and odd numbers:

Odd ... 21.363636

dtype: float64

4.1.2 Dictionary to Dataframe

Problem Statement:

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

Code:

```
import pandas as pd
```

```
# Provided dictionary of lists
```

```
data = {  
    'Name': ['Alice', 'Bob', 'Charlie'],  
    'Age': [25, 30, 35],  
}
```

```
# Convert the dictionary to a DataFrame
```

```
df = pd.DataFrame(data)
```

```
# Display the original DataFrame
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
# Adding a new
```

```
Name = input("New name: ")
```

```
Age = int(input("New age: "))
```

```
new = {'Name':Name, 'Age':Age}
```

```
df.loc[len(df)] = new
```

```
# Display the DataFrame after adding a new row
```

```
print("After adding a row:\n",df)
```

```
# Modifying a row
```

```
index = int(input("Index of row to modify: "))
```

```
age = int(input("New age: "))
```

```
df.at[index, 'Age']=age
```

```
# Display the DataFrame after modifying a row
```

```
print("After modifying a row:")
```

```
print(df)
```

```
# Deleting a row
```

```
delt = int(input("Index of row to delete: "))
```

```
df = df.drop(delt).reset_index(drop=True)
```

```
# Display the DataFrame after deleting a row
```

```
print("After deleting a row:")
```

```
print(df)
```

```
# Adding a new column
```

```
gen = input("Enter genders separated by space: ").split()
```

```
df['Gender']=gen
```

```
# Display the DataFrame after adding a new column
```

```
print("After adding a new column:")
```

```
print(df)
```

```
# Modifying a column
```

```
df['Name']=df['Name'].str.upper()
```

```
# Display the DataFrame after modifying a column
```

```
print("After modifying a column:")
```

```
print(df)
```

```
# Deleting a column
```

```
df = df.drop(columns = ['Age'])\
```

```
# Display the DataFrame after deleting a column
```

```
print("After deleting a column:")
print(df)
```

Output:

Average time
0.142 s
142.50 ms

Maximum time
0.157 s
157.00 ms

1 out of 1 shown test case(s) passed
1 out of 1 hidden test case(s) passed

Test case 1 157 ms Debug

Expected output

Original Dataframe:
.....Name Age
0.....Alice...25
1.....Bob...30
2.....Charlie...35
New name: Susan
New age: 29
After adding a row:
.....Name Age
0.....Alice...25
1.....Bob...30
2.....Charlie...35
3.....Susan...29
Index of row to modify: 0
New age: 27
After modifying a row:
.....Name Age
0.....Alice...27
1.....Bob...30
2.....Charlie...35
3.....Susan...29
Index of row to delete: 3
After deleting a row:
.....Name Age
0.....Alice...27
1.....Bob...30
2.....Charlie...35
Enter genders separated by space: F M M
After adding a new column:

Actual output
Original Dataframe:
.....Name Age
0.....Alice...25
1.....Bob...30
2.....Charlie...35
New name: Susan
New age: 29
After adding a row:
.....Name Age
0.....Alice...25
1.....Bob...30
2.....Charlie...35
3.....Susan...29
Index of row to modify: 0
New age: 27
After modifying a row:
.....Name Age
0.....Alice...27
1.....Bob...30
2.....Charlie...35
3.....Susan...29
Index of row to delete: 3
After deleting a row:
.....Name Age
0.....Alice...27
1.....Bob...30
2.....Charlie...35
Enter genders separated by space: F M M
After adding a new column:

Enter genders separated by space: F M M

After adding a new column:
.....Name Age Gender
0.....Alice...27.....F
1.....Bob...30.....M
2.....Charlie...35.....M
After modifying a column:
.....Name Age Gender
0.....ALICE...27.....F
1.....BOB...30.....M
2.....CHARLIE...35.....M
After deleting a column:
.....Name Gender
0.....ALICE.....F
1.....BOB.....M
2.....CHARLIE.....M

Enter genders separated by space: F M M

After adding a new column:
.....Name Age Gender
0.....Alice...27.....F
1.....Bob...30.....M
2.....Charlie...35.....M
After modifying a column:
.....Name Age Gender
0.....ALICE...27.....F
1.....BOB...30.....M
2.....CHARLIE...35.....M
After deleting a column:
.....Name Gender
0.....ALICE.....F
1.....BOB.....M
2.....CHARLIE.....M

4.1.3 Student Information

Problem Statement:

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.

- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Code:

```
import pandas as pd
# Read the text file into a DataFrame
file = input()
data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])
# write your code here..
print("First five rows:")
print(data.head())
average_age = round(data["Age"].mean(), 2)
print(f"Average age: {average_age}")
filtered_students = data[data["Grade"] <= "B"]
print("Students with a grade up to B")
print(filtered_students)
```

Output:

Average time
0.042 s
41.50 ms

Maximum time
0.049 s
49.00 ms

1 out of 1 shown test case(s) passed
1 out of 1 hidden test case(s) passed

Test case 1 49 ms
Debug

Expected output
Actual output

studentdata.txt
studentdata.txt

First five rows:
First five rows:

```

...Name Age Grade
0 John 25 A
1 Alin 22 B
2 Emma 24 A
3 Mike 23 C
4 Sony 21 B
Average age: 23.29
Students with a grade up to B:
...Name Age Grade
0 John 25 A
1 Alin 22 B
2 Emma 24 A
4 Sony 21 B
5 Amia 26 A

```

```

...Name Age Grade
0 John 25 A
1 Alin 22 B
2 Emma 24 A
3 Mike 23 C
4 Sony 21 B
Average age: 23.29
Students with a grade up to B:
...Name Age Grade
0 John 25 A
1 Alin 22 B
2 Emma 24 A
4 Sony 21 B
5 Amia 26 A

```

4.2.1 Month with the Highest Total Sales

Problem Statement:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.

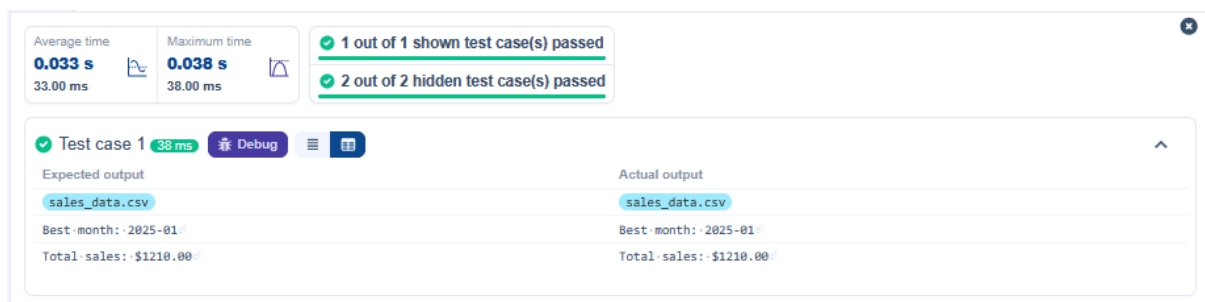
- Also, display the total sales for the best month.

Code:

```
import pandas as pd
# Prompt the user for the file name
file_name = input()
# Load the data
df = pd.read_csv(file_name)
df['Date'] = pd.to_datetime(df['Date'])
df['Total_Sales_Amount'] = df['Quantity'] * df['Price']
Monthly_Sales = df.groupby(df['Date'].dt.strftime('%Y-%m'))['Total_Sales_Amount'].sum()

# Find the month with the highest total sales
best_month = Monthly_Sales.idxmax()
highest_sales = Monthly_Sales.max()
print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")
```

Output:



The screenshot shows a code execution interface with the following details:

- Performance Metrics:**
 - Average time: 0.033 s (33.00 ms)
 - Maximum time: 0.038 s (38.00 ms)
- Test Results:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case 1 Details:**
 - Duration: 38 ms
 - Buttons: Debug, Run, Stop
- Expected vs Actual Output:**

Expected output	Actual output
sales_data.csv	sales_data.csv
Best month: 2025-01	Best month: 2025-01
Total sales: \$1210.00	Total sales: \$1210.00

4.2.2 Best Selling Product

Problem Statement:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Code:

```
import pandas as pd

# Prompt the user for the file name
file_name = input()
# Load the data
df = pd.read_csv(file_name)
product_sales = df.groupby("Product")["Quantity"].sum()
# Find the product with the highest total quantity sold
best_product = product_sales.idxmax()
highest_quantity = product_sales.max()
# Display the result
print(f"Best selling product: {best_product}")
print(f"Total quantity sold: {highest_quantity}")
```

Output:

The screenshot displays a code execution interface. At the top, performance metrics show an average time of 0.025 s (25.00 ms) and a maximum time of 0.028 s (28.00 ms). Test results indicate that 1 out of 1 shown test case(s) passed and 2 out of 2 hidden test case(s) passed. Below this, a section for 'Test case 1' (27 ms) shows a 'Debug' button and a list of icons. The 'Expected output' and 'Actual output' sections both display the same text: 'sales_data.csv', 'Best selling product: Product A', and 'Total quantity sold: 23'.

4.2.3 City that Sold the Most Products

Problem Statement:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

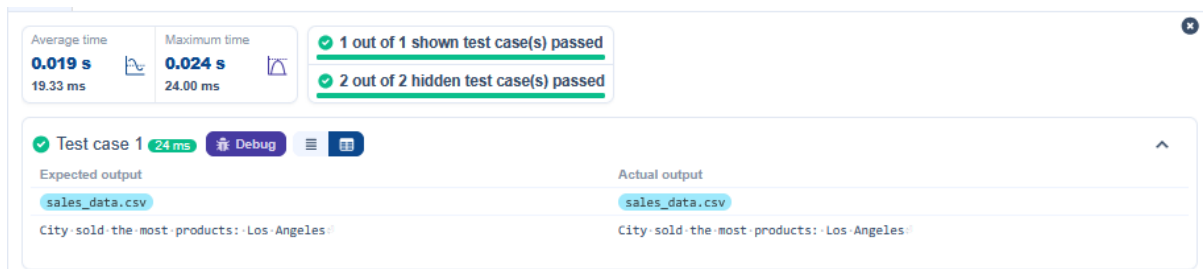
Code:

```
import pandas as pd

# Prompt the user for the file name
file_name = input()
# Load the data
df = pd.read_csv(file_name)
# write the code..
city_sales = df.groupby("City")["Quantity"].sum()
best_city = city_sales.idxmax()

# Display the result
print(f"City sold the most products: {best_city}")
```

Output:



The screenshot shows a code execution interface with the following details:

- Performance Metrics:**
 - Average time: 0.019 s (19.33 ms)
 - Maximum time: 0.024 s (24.00 ms)
- Test Results:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case 1:**
 - Status: Passed (24 ms)
 - Buttons: Debug, Run, Stop
- Expected output:**

```
sales_data.csv
City sold the most products: Los Angeles
```
- Actual output:**

```
sales_data.csv
City sold the most products: Los Angeles
```

4.2.4 Most Frequently Sold Product Pairs

Problem Statement:

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Code:

```
import pandas as pd
```

```

from itertools import combinations
from collections import Counter

# Prompt user to input the file name
file_name = input()

# Read data from the specified CSV file
df = pd.read_csv(file_name)

# write the code
grouped = df.groupby("Date")["Product"].apply(list)

pairs_counter = Counter()
for products in grouped:
    pairs = combinations(sorted(products), 2)
    pairs_counter.update(pairs)

max_frequency = max(pairs_counter.values())
most_common_pairs = [(pair, freq) for pair, freq in pairs_counter.items() if freq ==
max_frequency]

# Output the most frequent product pairs
for (product1, product2), frequency in most_common_pairs:
    print(f"{product1} and {product2}: {frequency} times")

```

Output:

Average time

0.020 s

19.67 ms

Maximum time

0.020 s

20.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

20 ms

Debug

Expected output

sales_data.csv

Product A and Product B: 2 times

Product A and Product C: 2 times

Actual output

sales_data.csv

Product A and Product B: 2 times

Product A and Product C: 2 times

4.2.5 Titanic Dataset Analysis and Data Cleaning

Problem Statement:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.

8. Fill missing values in the 'Embarked' column with the most frequent value (mode()).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

Code:

```
import pandas as pd
import numpy as np
```

```
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
print(data.head())
```

```
# 1. Display the first 5 rows of the dataset
print(data.tail())
```

```
# 2. Display the last 5 rows of the dataset
print(data.shape)
```

```
# 3. Get the shape of the dataset
print(data.info())
```

```
# 4. Get a summary of the dataset (info)
print(data.describe())
```

```
# 5. Get basic statistics of the dataset
print(data.isnull().sum())
```

```
# 6. Check for missing value
median_age = data['Age'].median()
data['Age'].fillna(median_age, inplace = True)
```

```
# 7. Fill missing values in the 'Age' column with the median age
mode_embarked = data['Embarked'].mode() [0]
data['Embarked'].fillna(mode_embarked, inplace = True)
```

```
# 8. Fill missing values in the 'Embarked' column with the mode
data.drop('Cabin', axis=1, inplace = True)
```

```
# 9. Drop the 'Cabin' column due to many missing values
data['FamilySize'] = data['SibSp'] + data['Parch']
```

```
# 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'
```

Output:

Average time
0.185 s
185.00 ms

Maximum time
0.185 s
185.00 ms

1 out of 1 shown test case(s) passed

Test case 1185 ms

Debug

Expected output

Actual output

```
... PassengerId Survived Pclass Fare Cabin Embarked
0 1 0 3 7.2500 NaN S
1 2 1 1 71.2833 C85 C
2 3 1 3 7.9250 NaN S
3 4 1 1 53.1000 C123 S
4 5 0 3 8.0500 NaN S
```

```
[5 rows x 12 columns]
```

```
... PassengerId Survived Pclass Fare Cabin Embarked
886 887 0 2 13.00 NaN S
887 888 1 1 30.00 B42 S
888 889 0 3 23.45 NaN S
889 890 1 1 30.00 C148 C
890 891 0 3 7.75 NaN Q
```

```
[5 rows x 12 columns]
```

```
(891, 12)
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 12 columns):
```

```
# Column Non-Null Count Dtype
PassengerId 891 non-null int64
Survived 891 non-null int64
Pclass 891 non-null int64
Name 891 non-null object
Sex 891 non-null object
Age 714 non-null float64
SibSp 891 non-null int64
Parch 891 non-null int64
Ticket 891 non-null object
Fare 891 non-null float64
Cabin 284 non-null object
```

```
... PassengerId Survived Pclass Fare Cabin Embarked
0 1 0 3 7.2500 NaN S
1 2 1 1 71.2833 C85 C
2 3 1 3 7.9250 NaN S
3 4 1 1 53.1000 C123 S
4 5 0 3 8.0500 NaN S
```

```
[5 rows x 12 columns]
```

```
... PassengerId Survived Pclass Fare Cabin Embarked
886 887 0 2 13.00 NaN S
887 888 1 1 30.00 B42 S
888 889 0 3 23.45 NaN S
889 890 1 1 30.00 C148 C
890 891 0 3 7.75 NaN Q
```

```
[5 rows x 12 columns]
```

```
(891, 12)
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 12 columns):
```

```
# Column Non-Null Count Dtype
PassengerId 891 non-null int64
Survived 891 non-null int64
Pclass 891 non-null int64
Name 891 non-null object
Sex 891 non-null object
Age 714 non-null float64
SibSp 891 non-null int64
Parch 891 non-null int64
Ticket 891 non-null object
Fare 891 non-null float64
Cabin 284 non-null object
```

Average time 0.185 s 185.00 ms Maximum time 0.185 s 185.00 ms 1 out of 1 shown test case(s) passed	
-10- Cabin 204 non-null object- -11- Embarked 889 non-null object- dtypes: float64(2), int64(5), object(5) memory usage: 66.2+ KB None PassengerId Survived Pclass SibSp Parch Fare count 891.000000 891.000000 891.000000 891.000000 891.000000 891.000000 mean 446.000000 0.383838 2.308642 0.523008 0.381594 32.204208 std 257.353842 0.486592 0.836071 1.102743 0.806057 49.693429 min 1.000000 0.000000 1.000000 0.000000 0.000000 0.000000 25% 223.500000 0.000000 2.000000 0.000000 0.000000 7.910400 50% 446.000000 0.000000 3.000000 0.000000 0.000000 14.454200 75% 668.500000 1.000000 3.000000 1.000000 0.000000 31.000000 max 891.000000 1.000000 3.000000 8.000000 6.000000 512.329200	-10- Cabin 204 non-null object- -11- Embarked 889 non-null object- dtypes: float64(2), int64(5), object(5) memory usage: 66.2+ KB None PassengerId Survived Pclass SibSp Parch Fare count 891.000000 891.000000 891.000000 891.000000 891.000000 891.000000 mean 446.000000 0.383838 2.308642 0.523008 0.381594 32.204208 std 257.353842 0.486592 0.836071 1.102743 0.806057 49.693429 min 1.000000 0.000000 1.000000 0.000000 0.000000 0.000000 25% 223.500000 0.000000 2.000000 0.000000 0.000000 7.910400 50% 446.000000 0.000000 3.000000 0.000000 0.000000 14.454200 75% 668.500000 1.000000 3.000000 1.000000 0.000000 31.000000 max 891.000000 1.000000 3.000000 8.000000 6.000000 512.329200
[8 rows x 7 columns] PassengerId 0 Survived 0 Pclass 0 Name 0 Sex 0 Age 177 SibSp 0 Parch 0 Ticket 0 Fare 0 Cabin 687 Embarked 2 dtype: int64	[8 rows x 7 columns] PassengerId 0 Survived 0 Pclass 0 Name 0 Sex 0 Age 177 SibSp 0 Parch 0 Ticket 0 Fare 0 Cabin 687 Embarked 2 dtype: int64

4.2.6 Titanic Dataset Analysis and Data Cleaning-2

Problem Statement:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

Code:

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']

# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
data['Alone'] = np.where(data['FamilySize'] == 0, 1, 0)

# 2. Convert 'Sex' to numeric (male: 0, female: 1)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

# 3. One-hot encode the 'Embarked' column
data = pd.get_dummies(data, columns = ['Embarked'], drop_first = True)

# 4. Get the mean age of passengers
mean_age = data['Age'].mean()
print(mean_age)

# 5. Get the median fare of passengers
median_fare = data['Fare'].median()
print(median_fare)

# 6. Get the number of passengers by class
passengers_by_class = data['Pclass'].value_counts()
```

```
print(passengers_by_class)
```

7. Get the number of passengers by gender

```
passengers_by_gender = data['Sex'].value_counts().sort_index()
print(passengers_by_gender)
```

8. Get the number of passengers by survival status

```
passengers_by_survival = data['Survived'].value_counts().sort_index()
print(passengers_by_survival)
```

9. Calculate the survival rate

```
survival_rate = data['Survived'].mean()
print(survival_rate)
```

10. Calculate the survival rate by gender

```
survival_rate_by_gender = data.groupby('Sex')['Survived'].mean()
print(survival_rate_by_gender)
```

Output:

Average time
0.022 s
22.00 ms

Maximum time
0.022 s
22.00 ms

1 out of 1 shown test case(s) passed

Test case 1 22 ms Debug

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3...491	3...491
1...216	1...216
2...184	2...184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0...577	0...577
1...314	1...314
Name: Sex, dtype: int64	Name: Sex, dtype: int64
0...549	0...549
1...342	1...342
Name: Survived, dtype: int64	Name: Survived, dtype: int64
0.3838383838383838	0.3838383838383838
Sex	Sex
0...0.188908	0...0.188908
1...0.742038	1...0.742038
Name: Survived, dtype: float64	Name: Survived, dtype: float64

4.2.7 Titanic Dataset Analysis and Data Cleaning-3

Problem Statement:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

Code:

```
import pandas as pd
import numpy as np
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# 1. Calculate the survival rate by class
print(data.groupby('Pclass')['Survived'].mean())
# 2. Calculate the survival rate by embarked location
print(data.groupby('Embarked_S')['Survived'].mean())
# 3. Calculate the survival rate by family size
print(data.groupby('FamilySize')['Survived'].mean())
# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())
# 5. Get the average fare by class
print(data.groupby('Pclass')['Fare'].mean())
# 6. Get the average age by class
print(data.groupby('Pclass')['Age'].mean())
# 7. Get the average age by survival status
print(data.groupby('Survived')['Age'].mean())
# 8. Get the average fare by survival status
print(data.groupby('Survived')['Fare'].mean())
# 9. Get the number of survivors by class
print(data[data['Survived'] == 1]['Pclass'].value_counts())
# 10. Get the number of non-survivors by class
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

Output:

Average time
0.057 s
57.00 ms

Maximum time
0.057 s
57.00 ms

1 out of 1 shown test case(s) passed

Test case 1 57 ms Debug

Expected output	Actual output
Pclass	Pclass
1...0.629638	1...0.629638
2...0.472826	2...0.472826
3...0.242363	3...0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0...0.586873	0...0.586873
1...0.336957	1...0.336957
Name: Survived, dtype: float64	Name: Survived, dtype: float64
FamilySize	FamilySize
0...0.303538	0...0.303538
1...0.552795	1...0.552795
2...0.578431	2...0.578431
3...0.724138	3...0.724138
4...0.200000	4...0.200000
5...0.136364	5...0.136364
6...0.333333	6...0.333333
7...0.000000	7...0.000000
10...0.000000	10...0.000000
Name: Survived, dtype: float64	Name: Survived, dtype: float64
IsAlone	IsAlone
0...0.505650	0...0.505650
1...0.303538	1...0.303538
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Pclass	Pclass
1...84.154687	1...84.154687
2...20.662183	2...20.662183
3...13.675550	3...13.675550
Name: Fare, dtype: float64	Name: Fare, dtype: float64
Pclass	Pclass
1...38.233441	1...38.233441
2...29.877630	2...29.877630
3...25.140620	3...25.140620

3...25.140620	3...25.140620
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0...30.626179	0...30.626179
1...28.343698	1...28.343698
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0...22.117887	0...22.117887
1...48.395408	1...48.395408
Name: Fare, dtype: float64	Name: Fare, dtype: float64
1...136	1...136
3...119	3...119
2...87	2...87
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
3...372	3...372
2...97	2...97
1...80	1...80
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64

4.2.8 Titanic Dataset Analysis and Data Cleaning-4

Problem Statement:

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- 1. Get the number of survivors by gender (Sex).

2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

Code:

```
import pandas as pd
import numpy as np
```

```
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
# 1. Get the number of survivors by gender
survivors_by_gender = data[data['Survived'] == 1]['Sex'].value_counts()
print(survivors_by_gender)
```

```
# 2. Get the number of non-survivors by gender
non_survivors_by_gender = data[data['Survived'] == 0]['Sex'].value_counts()
print(non_survivors_by_gender)
```

```
# 3. Get the number of survivors by embarked location
survivors_by_embarked_s = data[data['Survived'] == 1]['Embarked_S'].value_counts()
print(survivors_by_embarked_s)
```

```
# 4. Get the number of non-survivors by embarked location
non_survivors_by_embarked_s = data[data['Survived'] == 0]['Embarked_S'].value_counts()
print(non_survivors_by_embarked_s)
```

```
# 5. Calculate the percentage of children (Age < 18) who survived
children = data[data['Age'] < 18]
children_survival_rate = children['Survived'].mean()
print(children_survival_rate)
```

```
# 6. Calculate the percentage of adults (Age >= 18) who survived
adults = data[data['Age'] >= 18]
```

```
adults_survival_rate = adults['Survived'].mean()
print(adults_survival_rate)
```

7. Get the median age of survivors

```
median_age_survivors = data[data['Survived'] == 1]['Age'].median()
print(median_age_survivors)
```

8. Get the median age of non-survivors

```
median_age_non_survivors = data[data['Survived'] == 0]['Age'].median()
print(median_age_non_survivors)
```

9. Get the median fare of survivors

```
median_fare_survivors = data[data['Survived'] == 1]['Fare'].median()
print(median_fare_survivors)
```

10. Get the median fare of non-survivors

```
median_fare_non_survivors = data[data['Survived'] == 0]['Fare'].median()
print(median_fare_non_survivors)
```

Output:

Average time

0.030 s
30.00 ms



Maximum time

0.030 s
30.00 ms



1 out of 1 shown test case(s) passed



Test case 1 30 ms Debug



Expected output

Actual output

female...233

female...233

male...109

male...109

Name: Sex, dtype: int64

Name: Sex, dtype: int64

male...468

male...468

female...81

female...81

Name: Sex, dtype: int64

Name: Sex, dtype: int64

1...217

1...217

0...125

0...125

Name: Embarked_S, dtype: int64

Name: Embarked_S, dtype: int64

1...427

1...427

0...122

0...122

Name: Embarked_S, dtype: int64

Name: Embarked_S, dtype: int64

0.5398230088495575

0.5398230088495575

0.3810316139767055

0.3810316139767055

28.0

28.0

28.0

28.0

26.0

26.0

10.5

10.5